Course Overview

Title :	**ADVANCED MACHINE LEARNING**
Duration :	24 hours of content + 3h project support
Professor :	**Paul DUBOIS**
Email:	p.dubois@centralesupelec.fr
Website :	Edunao

Course description

This course is designed to give students a deeper understanding of the core theoretical concepts in machine learning. With a focus on the intersection of theory and applications, the course will emphasise essential aspects such as optimisation, high-dimensional data analysis, and statistical robustness. It covers various supervised and unsupervised learning techniques, associated challenges and their practical applications. Additionally, Python programming lab sessions will be used to illustrate these challenges in real-world data scenarios.

Structure

Each session consists of a 1.5-hour lecture and a 1.5-hour lab. The labs will include hands-on exercises and practical assignments using Python.

Content (the order is subject to change)

1. Reminders on Machine Learning and Probabilistic Classifiers
2. Robust regression
3. Logistic Regression and Support Vector Machines
4. Density-based and Hierarchical Clustering
5. Non-negative Matrix Factorization
6. Mixture models Fitting and Expectation Maximization Algorithm
7. Model Order Selection
8. Dimension Reduction and Data Visualisation with t-SNE

Learning Objectives

By the end of the course, the students will be able to:

- Understand the theory behind some advanced machine-learning tools
- Identigy the most appropriate algorithm(s) given a problem.

- Evaluate and compare machine learning algorithms for a particular task.
- Understand and handle the difficulties of real-world data.
- Implement some standard and advanced algorithms in Python
- Apply these algorithms to real datasets and analyse the results

Prerequisites

The students are expected to have basic knowledge of:

- Linear algebra (matrix inversion, eigenvectors and eigenvalues, …)
- Probabilities and statistics (random variable, probability density function, expectation, likelihood, posterior, prior, Bayes rule, ...)
- Foundations of machine learning
- Python

Reading material

All course material and supplementary resources will be available on the **Edunao** page. The instructor will share essential announcements through the Edunao forum, and appropriate resources will be added after each class.

James, G., Witten, D., Hastie, T. and Tibshirani, R. (2013) An Introduction to Statistical Learning, with Applications in R. Springer.

Hastie, T., Tibshirani, R. and Friedman, J. (2009) The Elements of Statistical Learning: Data Mining, Inference, and Prediction. Second edition. Springer.

C. M. Bishop (2006). Pattern Recognition and Machine Learning. Springer.

Evaluation

Project reports: Students must submit an individual project report. The report should be ~5 pages long on the topic of their choice. The project must involve at least one of the techniques seen during this course.

Grading:

Project report	100%